

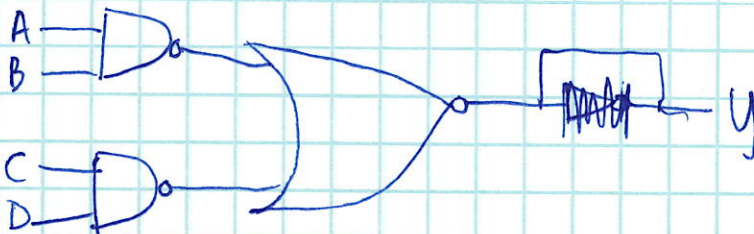
5c

Notes Pg 15

Implementation 1, no branching effort.

$$F = \frac{C_L}{C_{in0}} = \frac{1280 \text{ fF}}{10 \text{ fF}} = 128 \leftarrow \text{path electrical effort.}$$

Implementation 2 (corrected) ^{NOT}. No Branching outputs.



Path electrical effort: product of the effort per gate.

first gate 10 fF

last output 1280 fF

$$F = \frac{C_L}{C_{in0}} = \frac{1280 \text{ fF}}{10 \text{ fF}} = 128 ??$$

$$H = GFB$$

↑ gate ↑ path ↑ Branch

AWARE MISTAKE EXISTS - DON'T KNOW WHAT IS CORRECTION?

$$\text{Imp. 1: } H = 2 \times 128 \times 1 = 256$$

$$G = 2$$
$$F = 128$$
$$B = 1$$

$$N = \log_4(256) = 4.$$

Implementation 2:

$$G = \frac{80}{27} \quad F = 128$$
$$B = 1$$

$$H = \frac{80}{27} \times 128 = \frac{10240}{27} = 379.26$$

$$\log_4(H) \approx 4.284.$$

rounds to 4.

- (c) **Path effort and best number of stages.** For each implementation, compute the path electrical effort

$$F = \frac{C_L}{C_{in,0}}.$$

Assume no branching effort for Implementation 1. For Implementation 2, include any branching effort that appears on the path from A to Y . Then compute

$$H = GFB$$

and estimate the best number of stages using

$$N \approx \log_4 H.$$

Round N to a reasonable integer and justify your choice.

[4 points]

Please see Notes Pg 15 for
working & answer.

- (d) **Gate sizing.** For the implementation you believe is better, choose gate input capacitances so that each stage has approximately equal stage effort

$$h = H^{1/N}.$$

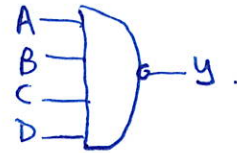
Work backward from the load and report the input capacitance of each stage on the path from A to Y. If you add output-buffer inverters, include them in the sizing. [5 points]

I believe that imp. 1 is better. ~~than the incorrect imp. 2.~~ Notes

Imp 1 N stages from 5c Page 15: $N = 4$. $H = 256$.

Individual stage effort $h = 256^{1/4} = 4$.

$$C_L = 1280 \text{ fF}$$



input gate capacitance = 10 fF.

4 stages within Imp 1 Nand gate? Am certain there is an error but. unclear where in my work & assumptions it is.

- (e) **Design-review recommendation.** Write a short design-review recommendation choosing between the two implementations. Your answer must discuss at least three of the following:

- logic correctness,
- number of stages,
- logical effort,
- parasitic delay,
- branching effort,
- input capacitance seen by signal A,
- robustness of using a large complex gate versus smaller gates.

↳ Not the corrected implementation, the original!

A purely qualitative answer such as "fewer stages is better" will receive limited credit. [3 points]

I would Recommend Imp. 1 as Imp. 2 is not logically correct (it isn't equivalent to Imp 1), contains fewer stages, less parasitic delay (just one gate!) has an increased robustness due to being a self-contained gate, and Requires less path-logical effort (2 versus Imp 2's $\frac{80}{27}$).

I have not used AI or LLMs, except for the occurrences where I have googled formulae and gotten to see ~~see~~ ^{scroll past} the LLM widget before the actual search results (wish it gave me an option to turn it off!).